Heart Disease Risk Prediction Report

April 6, 2026

Abstract

This report describes the development and evaluation of a heart disease risk prediction model using a UCI-style clinical dataset. The analysis includes data preprocessing, logistic regression modeling, and performance assessment with correlation, confusion matrix, and ROC visualizations.

1 Introduction

This project constructs a transparent heart disease classifier using logistic regression. The objective is to provide an interpretable clinical risk model with robust preprocessing and diagnostic evaluation.

2 Dataset

The dataset is loaded from `archive/heart_disease_uci.csv` and contains clinical features such as age, sex, chest pain type, resting blood pressure, cholesterol, maximum heart rate, ST depression, and thalassemia status. The response variable is encoded as a binary indicator for heart disease presence.

3 Data Preprocessing

Preprocessing includes:

- missing value imputation (mean for numeric features, mode for categorical features)
- normalization of the target label to binary form
- one-hot encoding of categorical predictors
- standard scaling of numeric variables
- train/test split with stratification to preserve class balance

4 Modeling

A logistic regression classifier is trained with the `liblinear` solver and `max_iter=1000`. The model is trained on 80% of the data and evaluated on the remaining 20%, using a stratified split to preserve outcome distribution.

5 Evaluation

Model performance is assessed with accuracy, a confusion matrix, and an ROC curve. The correlation heatmap provides insight into relationships among numeric features.

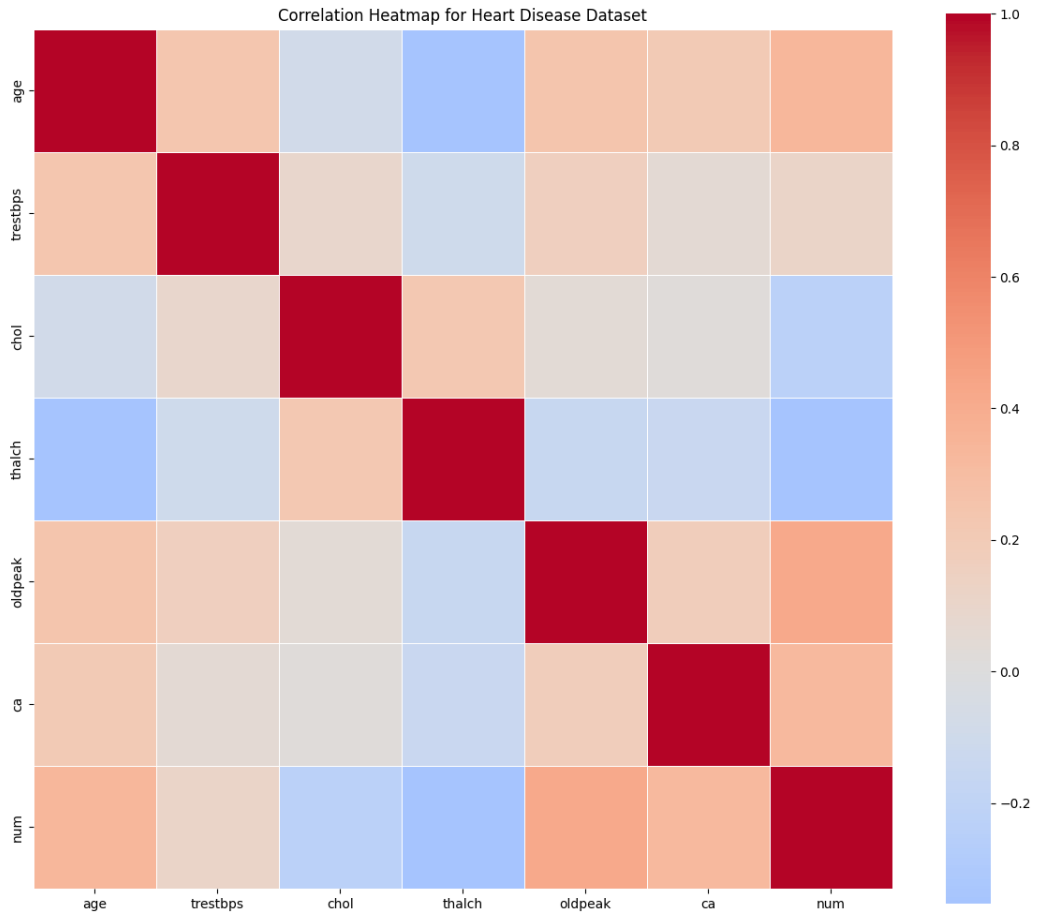

Figure 1: Correlation heatmap for numerical features in the heart disease dataset.

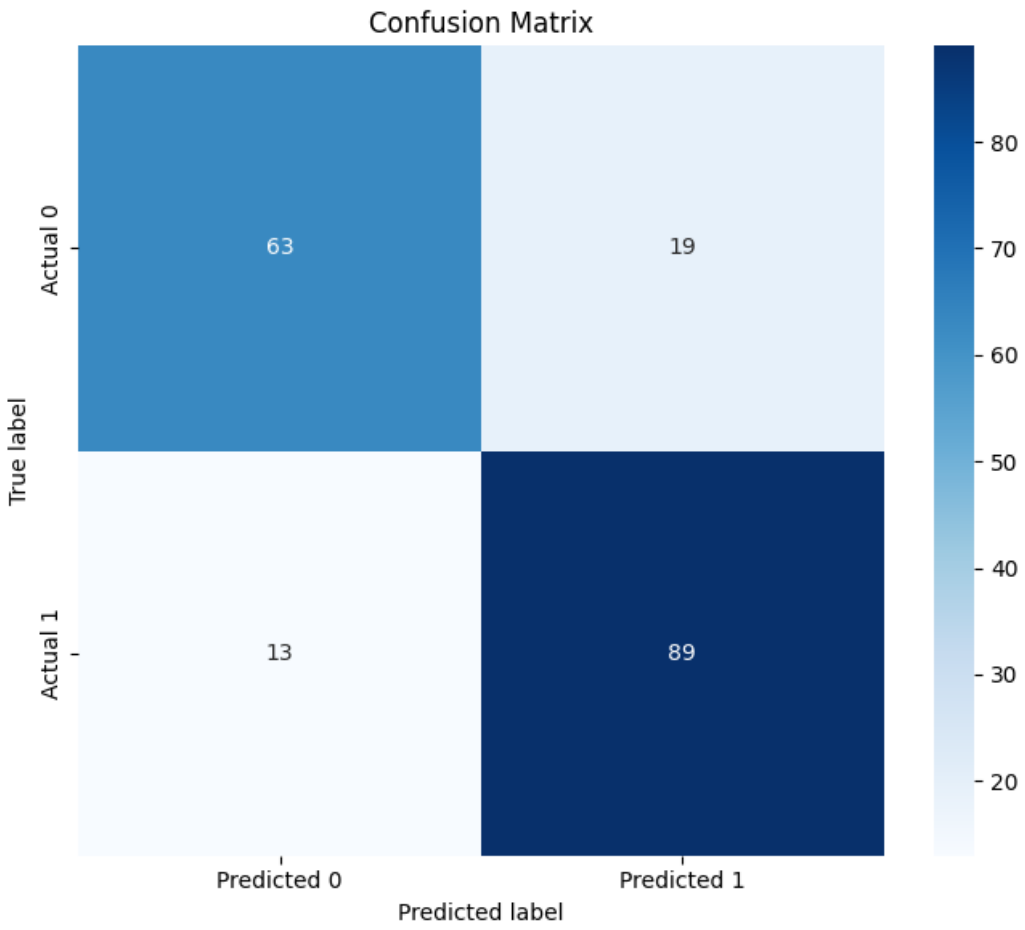

Figure 2: Confusion matrix for the logistic regression classifier on test data.

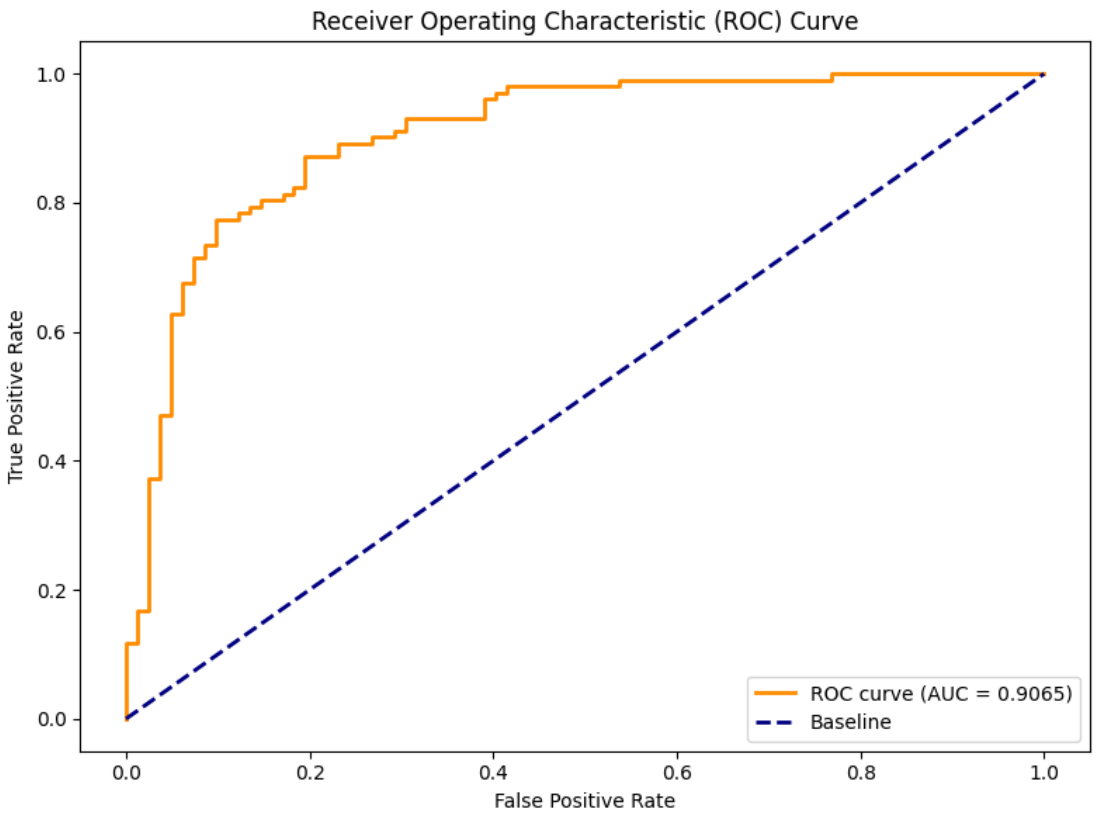

Figure 3: ROC curve for the trained logistic regression model.

6 Feature Importance

The model also produces a feature importance visualization showing the top predictors driving heart disease classification. This output is saved as `feature_importance.png` and helps explain which clinical variables are most influential for the logistic regression model.

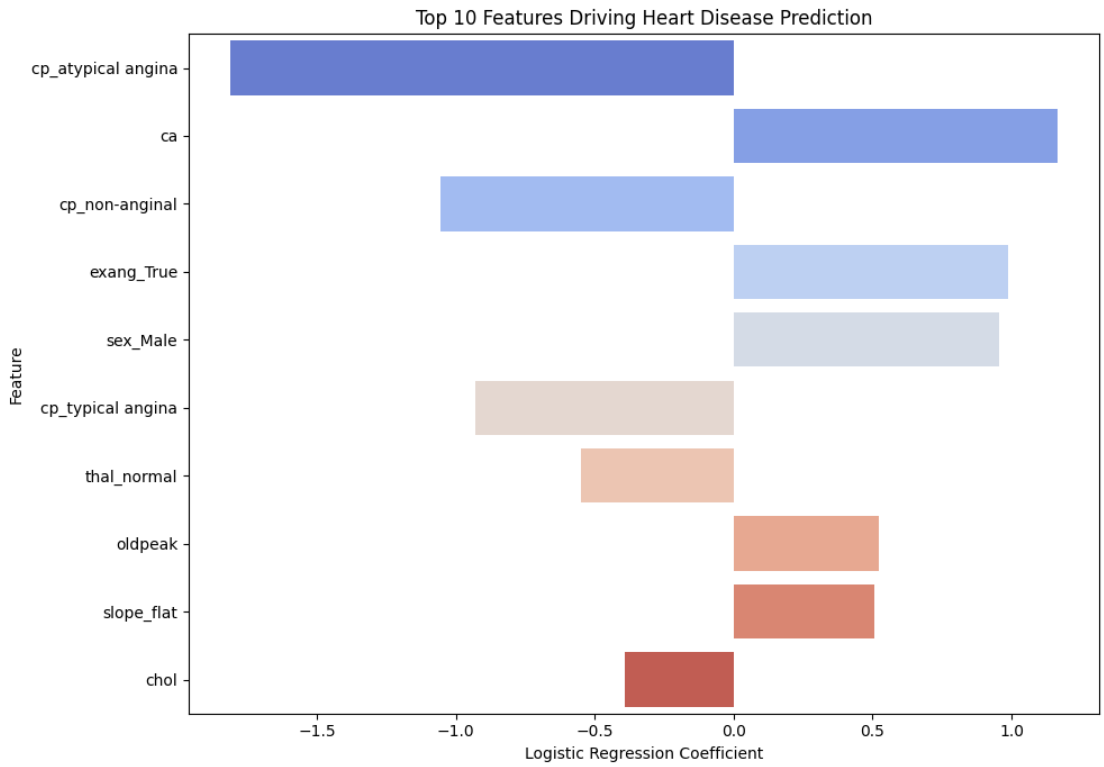

Figure 4: Top feature importance values from the logistic regression model.

7 Conclusion

The logistic regression pipeline provides a strong baseline for heart disease risk prediction, combining interpretable coefficients with high-level diagnostic plots. Future work can explore feature selection, alternate classifiers, and calibration for clinical risk scoring.